

# Image to Text Conversion Using Deep Learning Algorithms: Survey

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**Abstract**—Image Captioning (ICs) seamlessly combines the realms of Computer Vision (CV) and Natural Language Processing (NLP) task involved in producing textual sentences that summarise the image content in a way which is understandable to humans. The primary objective is to enable machines to convey the content of images in human-readable language. Image understanding and text formation are typically the two primary processes in the image captioning process. A family of Convolutional Neural Network (CNN) is used to retrieve visual information from an input image during the image understanding process. This visual information includes the image's scenes, relationships, and objects. To translate these visual features into coherent and contextually appropriate captions, recurrent neural network, long short-term memory, transformer-based models, etc. are used in the text generation step. In this article, we propose a survey about the previous research paper and the fundamental steps involved in image captioning tasks.

**Keywords**—Image, captioning, deep learning algorithm, language model, Text, Attention, word embedding, preprocessing.

## I. INTRODUCTION

Artificial Intelligence's (AIs) is widespread in today's world. Advancements in machine learning and deep learning, along with numerous methodologies, make AI an interdisciplinary science. As a result of changes in technology and the rise of mobile phones including images and data in the 21st century, there is a need for image caption generation.

Image Captioning (ICs) is a multidisciplinary topic that is gaining a lot of attention nowadays. Humans can accurately explain a vast amount of information about an image. Making computers understand the visual environment like humans has been a goal of AIs researchers. The main objective is to create intelligent systems that can produce coherent and appropriate captions in many scenarios for an image like a human described. Applying image captioning in the real world includes helping people who are blind while including audio for generated caption, boosting image search and retrieval, automating content generation for social media and news, and enhancing human-computer interaction in a variety of fields like robotics and autonomous vehicles, etc. For generating image descriptions, Computer Vision(CV) and

Natural Language Processing(NLP) domain is used[36 & 38]. Computer vision is a field of AI that enables the computer to understand and evaluate the visual elements in a video or image. A computer's ability to comprehend spoken and written language in a manner that is comparable to that of humans is referred to as NLP [37].

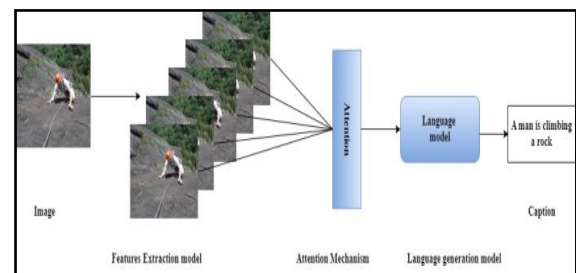


Fig. 1 Overview of image captioning based on attention mechanism

In ICs tasks, visual comprehension and text processing are the two fundamental concepts in image captioning. A family of CNN is used detect the visual features in an image and the language generation model RNN is used for generating a description for an image by utilizing detected features but it difficult to capture lengthy sequences. The objective of ICs is to produce a statement for an image that is true to the image's scene.

Traditionally, various approaches have been used in ICs, which include retrieval-based, template-based, Encoder-decoder based etc. These methods effectively align visual and textual information resulting in contextually relevant captions for diverse images. A template-based approach creates candidate sets from the image features. The candidate set's captions are ranked, and the caption with the greatest score is chosen as the latest output [37]. The retrieval-based method retrieves the most accurate description for an image from the pre-built sentence collection [10]. This method struggles to come up with an accurate description of a newly captured photo. Recent research has utilized an attention mechanism which allows the language model to create a caption by focusing on various areas of an input image thus increase coherent of a generated caption as shown in Fig. 1. Attention mechanism. Cognitive concept-driven and stimulus-driven attention mechanisms [2&3] are influenced by the human brain. Cognitive concept-driven, or Top-down attention, extracts

the high-level features from an image and it mainly focuses on the image region or an object that is most related to the current word. Stimulus-driven attention, or bottom-up attention, automatically captures the salient or important stimuli of our surroundings. It includes the involuntary and fast capture of attention by stimuli such as color, intensity, motion, and position.

## II. FUNDAMENTAL STEPS IN IMAGE CAPTIONING

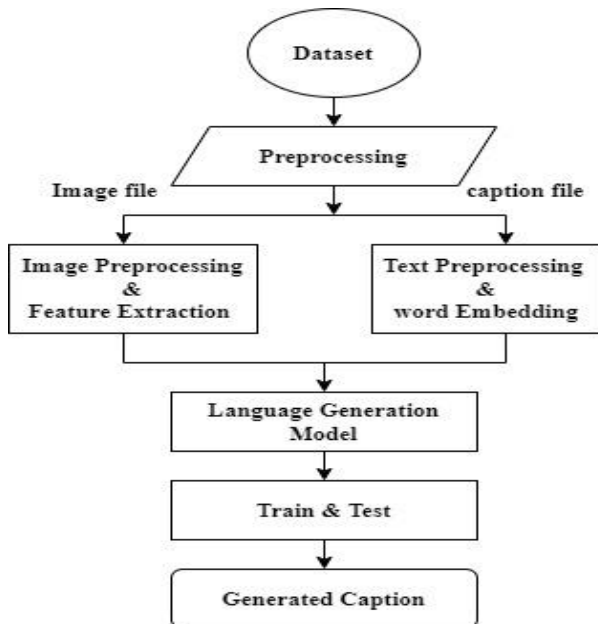


Fig. 2 Steps for implementing image captioning

### A. Dataset

Image captioning datasets usually include images and corresponding text descriptions. Flickr30k[28], MSCOCO[26], Flickr8k[27], Visual genome[30], SBU[29] are dataset used for ICs task.

### B. Image Preprocessing

Images should be resized to a constant resolution. Input images should normally be the same size for neural networks. Normalization is the conversion of pixel data to a standard range, frequently between 0 and 1 or -1 and 1. Else divide each pixel value by 255. It facilitates faster model convergence during training. Employ data augmentation techniques like random cropping, rotation, flipping, brightness alteration, and color jittering that helps to increase the diversity of the training data and improve model generalization. Fig. 2 shows the basic steps involved in performing the image captioning task.

### C. Feature Extraction

A pre-trained family of CNNs such as ResNet, VGG16, Inception, etc. are used to collect visual information in an image as shown in Fig. 3. These models trained on a large-scale dataset like ImageNet, which is capable of extracting visual representations and semantic information from images, can improve performance, decrease computation, and lengthen training time. Eliminate the fully connected layer for performing image captioning tasks by reviewing the recent research work. As a result, each image has a fixed-size feature vector.

### D. Text Preprocessing

Pre-processing the caption includes changing all the words to lowercase, eliminating punctuation marks, and short phrase with one character letter. Create a vocabulary by tokenizing the captions into words or subwords then map those words to distinct integer IDs by using in build tokenizer function present in keras. Add extra tokens called padding tokens, padding is used to guarantee that all captions are the same length.

### E. Building A Model

The output of the feature extraction model and pre-processed caption are given as the input for the language model. Language models like RNN, LSTM, GRU, or transformer-based models like BERT, GPT, etc. are used for predicting words for an image. Transformers fully replace RNNs with an attention mechanism. It more efficient since they process all data points simultaneously whereas RNN-based models handle data in a sequential manner.

### F. Train Test And Validation

The dataset contains both image and corresponding caption file, split the dataset according to system requirements.

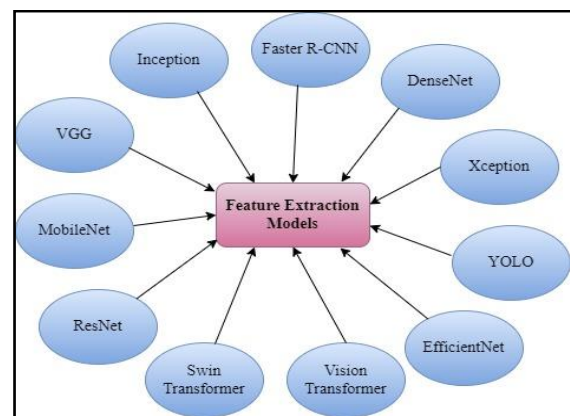


Fig. 3 Feature extraction model

TABLE I. POPULAR IMAGE CAPTIONING DATASET

| Dataset        | Total no of images | No.of captions per image | Total no of captions | Accuracy (CIDEr) |
|----------------|--------------------|--------------------------|----------------------|------------------|
| Flickr8k[17]   | 8092               | 5                        | 40,460               | 24.5             |
| Flickr30k [17] | 31,783             | 5                        | 1,58,915             | 67.3             |
| MSCOC O[17]    | 1,23,287           | 5                        | 6,16,435             | 124.2            |

Table 1 shows the various datasets with a total number of images and captions. It has both an image and a captions file. MSCOCO, Flickr30k and Flickr8k datasets are widely used by many researchers. Among the three datasets, Flickr8k gives a 24.5 CIDEr result. The Flickr30k dataset has a 67.3 CIDEr result. Finally, the MSCOCO dataset gives a 124.3 CIDEr result compared to Flickr8k and Flickr30k.

数据集：图像及文本描述 预处理：恒定分辨率，归一化，图像数据增强  
 特征提取：CNN模型，提取视觉表达以及语义信息，减少计算量，提高性能，延长训练时间  
 文本预处理：单词小写，创建词汇表，字幕单词占位相同

## III. LITERATURE SURVEY

The author present a parallel structure that exists between the image and the generated caption [1]. Feature vectors are formed by using ResNet-152, and extracted scene-specific features are given to Long Short Term Memory (LSTM) for generating words for scenes.

The author integrates **Bottom-Up (BU) and Top-Down (TU)** attention mechanisms for accurate Image captioning (ICs) [2]. Implemented BU attention using Faster RCNN [22] to find and localize the object in a picture and then concatenate with TD attention mechanisms to predict a task-specific context over the image region. The Visual Question Answering (VQA) model is implemented using BU attention features. Later, a similar method was adopted by [3].

ICs task still facing difficulties in describing an image containing a complex background. Ding et al. [4] introduced a unique ICs model for higher-level image features. This work combines lower-level image features (colorfulness, sharpness, and contrast) with higher-level image features (face and object recognition) to find a region of BU for generating captions more accurately.

This work utilized a visually enhanced gLSTM model as a language model to eliminate the gradient diminishing problem in guiding LSTM (gLSTM) [5]. Visual-enhanced gLSTM model captures semantic regions in an image which is similar human point of view in comparison with the original gLSTM model. Introduced Attention of Attention (AOA) an extension of Convolutional attention [6]. To address the issue of irrelevant attention, propose the Attention on Attention (AoA) module as a supplement to traditional attention processes. This method improves the encoder's aesthetics properties.

A semi-supervised system for image caption training has been suggested. They create a new semi-supervised learning technique using a novel GAN discriminator to make it easier to train on unpaired data [7]. Better comprehension of text and image is achieved by the suggested Multi-dense attention method, which also records salient information. [8]. Linear attention block was introduced [9]. Bilinear pooling can be fully utilized by X-linear attention to executing multi-modal reasoning while selectively utilizing visual input. The author introduced EnsCaption models composed of retrieval and generation-based image caption methods for better description [10].

Transformer-based architecture is proposed [11]. A stack of decoder layers and a stack of memory-enhanced encoding levels comprise the architecture. A meshed-memory transformer enhances caption by combining with a Transformer, acquiring multi-level representation, using acquired features and using lower- and higher-level picture information while captioning. Proposed a Multimodal transformer that incorporates both self-attention and co-attention for image captioning which captured complex scenes in the image [12].

This work presents Multilevel Similarity guided semantic matching to eliminate the dissimilarity between image and caption [13]. It contains a caption generator model that generates a caption and Multi-Level Similarity Fusion Network (MSFN) contains local and global semantic

similarity. Reinforcement learning techniques are used to combine local and global similarities. The above all work aimed to increase the accuracy of the generated caption, but it failed to describe the relationship between a human and an item in an image.

This work introduced the Human-Centric Captioning Model (HCCM), which attempts to describe human behavior in an image [14]. HCCM first focuses on human-centric feature hierarchization in an image, such as human features, background features, and activity-related features. This model describe the relationship between the human and object in an image. They created a Human-Centric COCO (HC-COCO) dataset with 78,462 phrases and 16,125 pictures, most of the captions describe human behaviour.

Attention-based encoder-decoder architecture was proposed [15]. Image features are detected by CNN (Xception) [24] is combined with object characteristics taken from YOLOv4 [23] is used in an Encoder part and GRU [25] is used as a language generation model in the decoder part. The Bahdanau soft attention system is employed in this research to improve machine translation. The overall smoothness and differentiability of the model are a result of its deterministic attention mechanism.

The author suggested Hybrid Attention Network (HAN) combination of human-caption and machine-caption attention mechanisms [16]. Adaptive fusion is used to concatenate both human-caption attention and machine attention, i.e., BU attention is used for reassigning the weight, detecting object semantics in an image even though it generates a less accurate caption but avoids the object hallucination, and TD attention improves the quality of generated sentences.

Wavelet transforms-based Convolutional Neural Networks (WCNN) to extract visual feature maps [17]. The Visual Attention-Prediction Network (VAPNs) combines spatial and channel focus to obtain semantic features. Refined semantic and context spatial links are retrieved using Contextual Spatial Relation Extractor (CSE) and VAPNs. LSTM is used for word prediction. Table 2 shows a comparison of image captioning methodologies.

The author presents a Relational Convergent Transformer (RCTs), an encoder part that contains a Relational Fusion Module (RFM's) function to extract both local and global semantic data, and Relational Convergent Attention (RCA), which uses global relational data in every transformer encoder layer to process the self-attention values to identify the contextual region [18]. To train a balanced captioner, the authors suggest a new Task-aware Decoupled Learning and Fusion (TDLF) technique [19]. The author proposed a Scene Graph with Transformer may infer reasonable and human-like sentences by utilizing the rich information and relational scenarios in a picture [20].

The author introduced a consensus-aware, vision-enhanced Transformer. Faster R-CNN creates a vision-enhanced encoder by extracting object regions from a picture and feeding them to a visual perception and memory-based attention module. [21]. Additionally, a graph convolutional network uses consensus information and updates it via the knowledge-based control module. The

captions are finally created by the decoder as a consensus-aware, which interacts cross-modally with semantic and visual information.

TABLE II.COMPARISON OF METHODOLOGIES USED FOR IMAGE CAPTION GENERATION

| Author              | Dataset used                             | Feature extraction model | Visual model             | Language generation model | Advantage and CIDEr score                                                                                                                                                           |
|---------------------|------------------------------------------|--------------------------|--------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fu et al. 2017      | Flickr8k, Flickr30k, MSCOCO              | ResNet-152               | Attention based          | LSTM                      | Scene-specific context is used to detect high-level features in an image and a specific scene-context language model is included for word generation. CIDEr score on MSCOCO is 95.5 |
| Anderson et al.2018 | MSCOCO, Visual genome                    | Faster R-CNN (ResNet101) | Attention based          | LSTM                      | Incorporate TD and BU attention to estimate the salient region and level of an object in an image<br>CIDEr score on MSCOCO is 95.5                                                  |
| Ding et al. 2019    | Flickr8k, Flickr30k, MSCOCO PASCAL, SBU. | VGG16-based Faster R-CNN | Bottom Up (BU) attention | LSTM                      | A BU attention model based on picture features is proposed, similar to how humans pay attention.<br>CIDEr score on MSCOCO is 1.103                                                  |
| Ding et al. 2019    | Flickr8k, Flickr30k, MSCOCO              | CNN (VGG-19)             | Attention based          | LSTM                      | Appropriately characterize the characteristics of humans in a picture is demonstrated by the attention mechanism.<br>CIDEr score on MSCOCO is 1.041                                 |
| Zhang et al. 2019   | Flickr8k                                 | CNN                      | Visual enhanced          | Visual enhance gLSTM      | Region and entire image-based are two visually enhanced methods are proposed. CIDEr score is 54.6.                                                                                  |
| Huang et al. 2019   | MSCOCO                                   | CNN                      | Self Attention           | LSTM                      | To address the issue of irrelevant attention, propose the Attention on Attention (AoA) module as a supplement to traditional attention processes.<br>CIDEr score is 129.8           |
| Kim et al. 2019     | scarcely-paired COCO                     | Faster R-CNN             | Top Down (TD) attention  | LSTM                      | Using a novel GAN discriminator, they develop a new semi-supervised learning strategy to facilitate training with unpaired data.<br>CIDEr score is 125.5                            |
| Wang et al.2019     | Flickr8k, Flickr30k, MSCOCO              | Faster R-CNN             | Dense Attention based    | LSTM                      | The proposed Multi-dense attention captures salient information and it has a better understanding of text and images.<br>CIDEr score on MSCOCO is 118.3                             |
| Pan et al. 2020     | MSCOCO                                   | Faster-RCNN              | X-Linear                 | Transformer               | Bilinear pooling can be fully utilized by X-Linear attention to execute multi-modal reasoning while selectively utilizing visual input.<br>CIDEr score on MSCOCO is 132.0           |
| Yang et al. 2020    | Flickr30k, MSCOCO                        | ResNet-101               | Top Down                 | LSTM                      | Introduced an EnsCaption model composed of retrieval and generation-based image caption methods for better description.<br>CIDEr score on MSCOCO is 125.5                           |
| Cornia et al. 2020  | COCO                                     | Faster R-CNN             | Self-Attention           | Transformer               | Meshed-memory transformer enhances caption generation and<br>CIDEr score is 131.2                                                                                                   |
| Yu et al.2020       | MSCOCO                                   | Faster-RCNN              | Attention based          | LSTM                      | Proposed a Multimodal transformer that incorporates both self- and co-attention which captures complex scenes in the image.<br>CIDEr score on MSCOCO is 130.0                       |



|                          |                              |              |                                     |             |                                                                                                                                                                                                       |
|--------------------------|------------------------------|--------------|-------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Li et al. 2021           | MSCOCO                       | Faster R-CNN | Attention based                     | LSTM        | Present a Multi-level similarity-guided semantic matching to eliminate the dissimilarity between image and caption. CIDEr score on MSCOCO is 128.5                                                    |
| Yang et al. 2022         | Human-Centric COCO (HC-COCO) | Faster-RCNN  | Attention based                     | LSTM        | The proposed model describes an image with an accurate caption and CIDEr on the HC-COCO dataset is 128.5                                                                                              |
| Al-Malla et al. 2022     | Flickr30k, MSCOCO            | CNN ,YOLO v4 | Attention (Bahdanau soft)           | GRU         | Proposed importance, which emphasizes front bigger objects in an image over behind little ones. CIDEr score on MSCOCO is 15.04                                                                        |
| Jiang et al. 2022        | Flickr30k, MSCOCO            | Faster R-CNN | Human attention + Machine Attention | LSTM        | BU attention reduces the problem of “object hallucination”. CIDEr score on MSCOCO is 124.3                                                                                                            |
| Sasibhooshan et al. 2023 | Flickr8k, Flickr30k, MSCOCO  | WCNN         | VAPN+CSE                            | LSTM        | This research work employs wavelet transforms based on Convolutional Neural Networks (WCNN) for extracting the semantic, spectral, and spatial features from an image. CIDEr score on MSCOCO is 124.2 |
| Chen et al. 2023         | MSCOCO                       | Faster R-CNN | Self attention                      | Transformer | This paper presents the Relational Convergent Transformer (RCTs) to accurately capture the relationship between image features. RCT generates the caption precisely and CIDEr score is 131.4          |
| Ding et al. 2023         | MSCOCO                       | Faster R-CNN | BU & TD                             | LSTM        | To train a balanced captioner, the authors suggest a new Task-aware Decoupled Learning and Fusion (TDLF) technique. CIDEr score on MSCOCO is 97.2                                                     |
| Li et al. 2023           | Flickr30k, MSCOCO            | Faster R-CNN | Attention based                     | Transformer | The proposed Scene Graph with Transformer may infer reasonable and human-like sentences by utilizing the rich information and relational scenarios in a picture. CIDEr score on Flickr30k is 78.5     |

## IV. EVALUATION METRICS

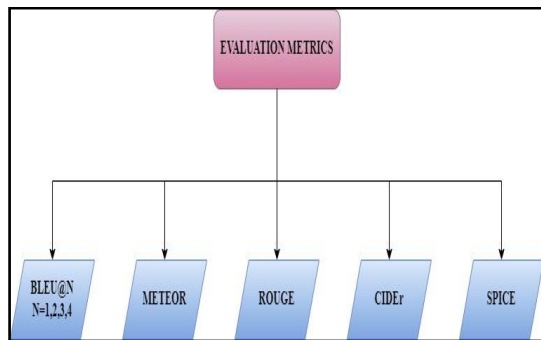


Fig. 4 Metrics for evaluating generated caption.

BLEU, METEOR, ROUGE, CIDEr, SPICE [35] are commonly used evaluation metrics as shown in Fig. 4.

## A. BLEU

Bilingual Evaluation Understudy is a popular and expensive method used to evaluate how closely a candidate sentence is matched with a reference sentence [31]. It uses n-gram precision ranging from 1 to 4 sentences. It scores ranges from 0 to 1.

Let  $P_n$  be the modified precision score for the whole text corpus, A - candidate sentence, B - reference sentence.

To compute

$$p_n = \frac{\sum_{A \in \{candidate\}n-grams} \sum_{B \in \{reference\}n-grams} count_{clip}(n-grams)}{\sum_{A \in \{candidate\}n-grams} count(n-grams)} \quad (1)$$

## B. ROUGE

Recall-Oriented Understudy for Gisting Evaluation is used to assess the similarities between the candidate and reference description by using various n-grams based on precision and recall score [32].

$$N = \frac{\sum_{B \in \text{reference sentences}} \sum_{gram_n \in B} c_{match}(gram_n)}{\sum_{B \in \text{reference sentences}} \sum_{gram_n \in B} c(gram_n)} \quad (2)$$

Where B - candidate description, N denote ROUGE - N, n - length of n-gram,  $c_{match}(gram_n)$  - maximum number of n-grams co-occur in collection of reference and candidate descriptions.

## C. CIDEr

Consensus-based Image Description Evaluation evaluates the created caption's quality based on how closely it matches the descriptions provided by human annotators. Its highly correlated with human consensus score and gives more weight-age to important n-gram [34].

指标：BLEU：评估候选与参考句子匹配程度（0-1） ROUGE：候选描述与参考描述匹配程度 CIDEr：创建标题的匹配程度 METEOR：候选词与参考词匹配程度

$$CIDEr_n(A, B) = \frac{1}{M} \sum_{i=1}^M \frac{g^n(A) \cdot g^n(B)}{\|g^n(A)\| \|g^n(B)\|} \quad (3)$$

Where A- candidate caption, B- reference caption, M- number reference caption,  $g^n(A)$ ,  $g^n(B)$ - n-gram in A, B formed by Term Frequency Inverse Document Frequency (TFIDF).

#### D. METEOR

Metric for evaluation of translation with explicit Ordering measures the word-for-word matching among the candidate and the reference description [33]. It calculates the recall, precision scores and then performs F1 harmonic-mean.

Where P = unigram precision, R = unigram recall, A = candidate caption, B = reference caption, chunks = collection of unigrams adjacent in candidate and reference caption.  $F_{mean}$  computed by using precision and recall via a harmonic mean

$$F_{mean} = \frac{10PR}{R+9P} \quad (3)$$

Computing penalty,

$$penalty = 0.5 * \left( \frac{\#chunks}{\#matched\_chunks} \right)^3 \quad (4)$$

Then, METEOR score computed as

$$score = F_{mean} * (1 - penalty) \quad (5)$$

#### V. CONCLUSION

ICs have drawn a large number of scholars in recent years. The combination of Natural Language Processing and Computer Vision contributed to this field's amazing evolution. This paper summarizes image captioning based on attention mechanisms. The attention mechanism directs the language model, which prioritizes semantic information, to appropriately construct a caption. First, we shortly discuss the early ICs approaches. Secondly, we listed out the fundamental steps for image caption generation task. Then this paper summarized existing research work. Finally, elaborate on the evaluation metrics. Future work will be implemented by using attention along with advanced language generation model in order to enhance the quality of generated caption.

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